

Dr Loizos Pelecanos

MEng (Hons) ACGI PhD DIC FHEA MIMarEST

Department of Architecture & Civil Engineering, University of Bath
Civil Engineer & Academic Researcher

Computational geotechnics · earthquake engineering · structural dynamics · offshore & marine infrastructure · sensing · optimisation · AI/ML



37 published journal articles	90+ research outputs	5 completed PhDs	59 MEng/MSc dissertations
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Research profile

Research combines computational mechanics, finite-element analysis, field monitoring and experimental evidence with optimisation and data-driven methods. Applications include foundations, buildings, bridges, dams, offshore wind infrastructure, marine systems and low-carbon engineering.

Core themes

- Computational geotechnics & soil-structure interaction
- Earthquake engineering & structural dynamics
- Offshore & marine infrastructure
- Infrastructure monitoring & distributed fibre-optic sensing
- Sustainable foundation engineering & optimisation
- AI/ML, inverse analysis & model updating

Academic trajectory

2016 — Appointed Lecturer in Civil Engineering, University of Bath
2013–2016 — Research Associate, University of Cambridge
2013 — PhD, Computational Earthquake Engineering, Imperial College London
2009 — MEng (Hons), Civil Engineering, Imperial College London

Academic & professional service

Editorial Boards: Computers & Geotechnics; Scientific Reports
Standards: Cyprus National Representative, CEN/TC 250/SC10/WG04
Dams: UK BDS Representative to ICOLD Technical Committee A
Professional: Member, Institute of Marine Engineering, Science & Technology (MIMarEST)
Teaching: Fellow of the Higher Education Academy (FHEA)

Research & supervision

Five completed PhD students and 59 completed MEng/MSc dissertations. Research and PhD-studentship funding exceeds £265k, spanning resilient infrastructure, monitoring, offshore energy and low-carbon foundation engineering.

Selected publications

Representative peer-reviewed work across the main research themes.

2026	Yang, Y., Shin, G.-B., Pelecanos, L. , Wang, C. C., Luo, L. & Soga, K. Inferring pile capacity using distributed strain sensing with PDE-constrained optimization. <i>Computers and Geotechnics</i> 196, 108131.
2026	Shin, G.-B., Soga, K., Yang, Y., Pelecanos, L. , Mevoli, F. A. & Yeskoo, A. Distributed fiber optic sensing for monitoring piles under eccentric load. <i>ASCE Journal of Geotechnical and Geoenvironmental Engineering</i> 152(9), 04026089.
2024	Williams, S., Pelecanos, L. & Darby, A. A practical two parameter model of pile-soil gapping for prediction of monopile offshore wind turbine dynamics. <i>Géotechnique</i> 74(3), 265-280.
2024	Abushama, K., Hawkins, W., Pelecanos, L. & Ibell, T. Embodied carbon optimisation of concrete pile foundations and comparison of the performance of different pile geometries. <i>Engineering Structures</i> 310, 118109.
2022	Williams, S., Pelecanos, L. & Darby, A. A Winkler model of monopile ratcheting under long-term dynamic cyclic loading. <i>Ocean Engineering</i> 266, 112625.
2018	Pelecanos, L. , Soga, K., Elshafie, M. Z. E. B., De Battista, N., Gue, C. Y., Ouyang, Y. & Seo, H. J. Distributed fibre optic sensing of axially loaded bored piles. <i>ASCE Journal of Geotechnical and Geoenvironmental Engineering</i> 144(3), 04017122.
2015	Pelecanos, L. , Kontoe, S. & Zdravkovic, L. A case study on the seismic performance of earth dams. <i>Géotechnique</i> 65(11), 923-935.
2013	Pelecanos, L. , Kontoe, S. & Zdravkovic, L. Numerical modelling of hydrodynamic pressures on dams. <i>Computers and Geotechnics</i> 53, 68-82.

Research areas & collaboration

Current research directions include nonlinear soil-foundation-structure interaction for offshore wind, hydrodynamic and marine infrastructure loading, seismic monitoring and model updating, distributed sensing for piles and foundations, low-carbon infrastructure optimisation, and AI-enabled computational engineering.

Online profiles

Website: <https://lpelecanos.github.io/>

University of Bath: <https://researchportal.bath.ac.uk/en/persons/loizos-pelecanos/>

ORCID: <https://orcid.org/0000-0001-6183-1439>

Google Scholar: <https://scholar.google.com/citations?user=rxJERUEAAAAJ&hl=en>

YouTube: <https://www.youtube.com/@LoizosPelecanosEngineering>

Email: L.Pelecanos@bath.ac.uk